Landslide Early Warning System Using Machine Learning

Le Tuan Khang, Duong Tuan Dung, Dinh Duc Quang, Tran An Phu, Luu Viet Anh

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Abstract

This paper presents a low-cost, real-time landslide early warning system designed for mountainous regions of Vietnam. The system integrates environmental sensors (soil moisture, ground vibration, pore-water pressure, rainfall), an ESP32 microcon- troller, and a Random Forest classification model to estimate landslide risk across five levels. Initial experiments show that the model achieves up to 99% accuracy on field- collected datasets, while the hardware maintains stable operation and reliable wireless communication. These results indicate that the system has potential to serve as an affordable disaster-warning solution suitable for community-level deployment.

1 Introduction

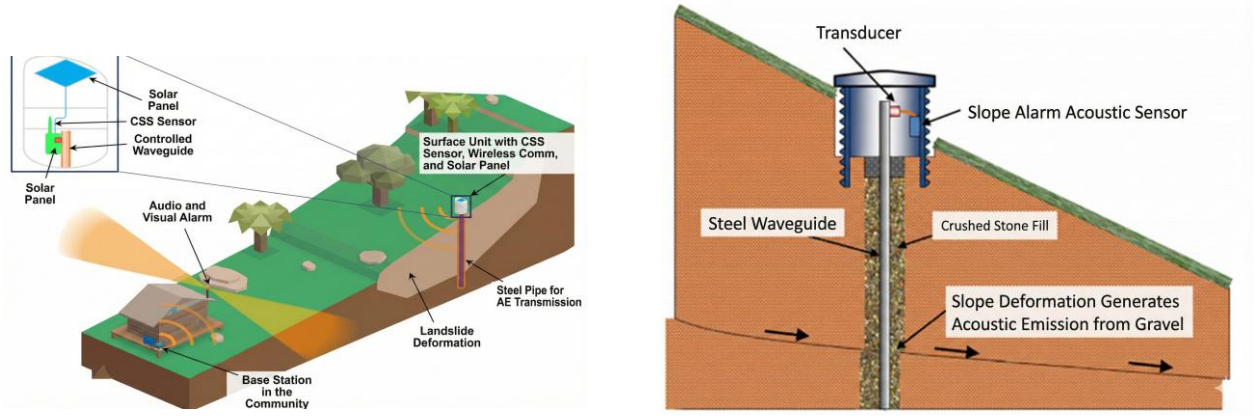
Landslides are among the most damaging natural hazards in Vietnam's mountainous regions, where steep terrain, heavy rainfall, and unstable geological conditions are prevalent. Climate change and increased human activities in vulnerable areas further contribute to the rising frequency and severity of landslides. Although several early warning systems have been studied and deployed, existing solutions are often costly, require specialized equipment, and demand complex technical infrastructure, limiting their applicability in rural communities. In this context, this study develops a low-cost, real-time landslide early warning system based on a combination of environmental sensors and a Random Forest model to esti- mate landslide risk levels. The system aims to provide a feasible technological platform for community-based disaster warning, featuring simple design, low power consumption, and easy scalability.



Figure 1: Landslides in Vietnam

2 System Design

The system consists of three main modules: (1) Sensor Unit, (2) Processing Unit, and (3) Warning and Communication Unit. This modular architecture facilitates installation, maintenance, and flexible expansion under real-world deployment conditions.



(a) System architecture (b) System components

Figure 2: System architecture and components

2.1 Sensor Unit

The sensor unit employs low-cost and reliable devices suitable for environmental data collection:

- MPU6050 accelerometer for detecting small-scale ground motion.
- FC-28 capacitive soil moisture sensor to measure soil water content (0–100%).
- Pore-water pressure sensor (kPa).

- Rain gauge measuring 24-hour accumulated rainfall.

Sensors are installed at a depth of 30–50 cm to reduce surface noise while capturing variations within the active soil layer. All sensors are IP65-rated for water resistance. The measurement interval is set to 5 minutes to balance data resolution with power consumption.

2.2 Processing Unit

The ESP32 microcontroller (240 MHz, 520 KB SRAM) is selected due to its low energy consumption, built-in Wi-Fi/Bluetooth, and ability to execute embedded machine learning models. The ESP32 performs:

- Sensor data acquisition
- Preprocessing (noise filtering, basic calibration)
- Random Forest inference
- Risk level estimation (1–5)
- Alert activation when high-risk levels (4–5) are detected

The ESP32 is mounted on a custom PCB with protective circuitry and stable power regulation.

2.3 Warning and Communication Unit

The system supports two types of warning:

- Local alerts via a 120 dB siren and high-intensity LED.
- Remote alerts via a SIM7680 GSM module sending SMS and HTTP data when high risk is detected.

The communication mechanism includes automatic retry logic and 2G support to improve reliability in areas with limited infrastructure.

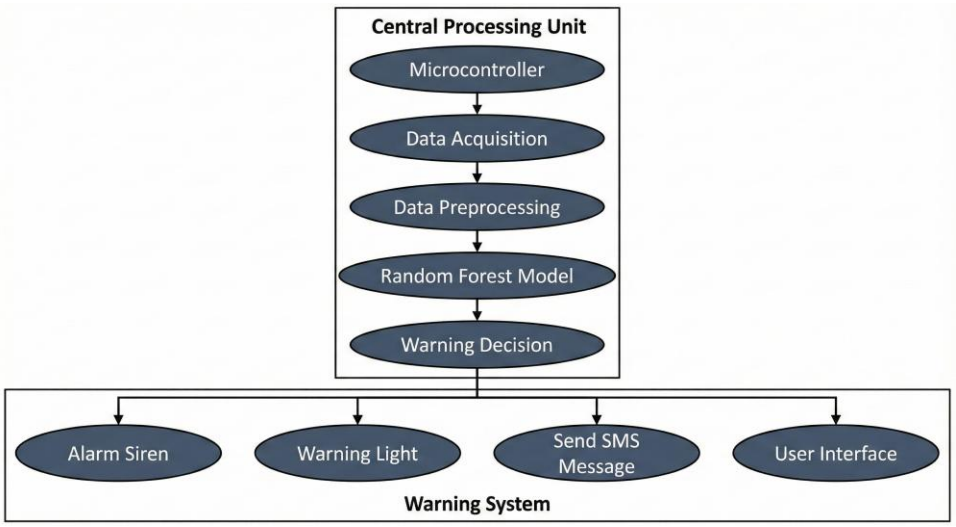


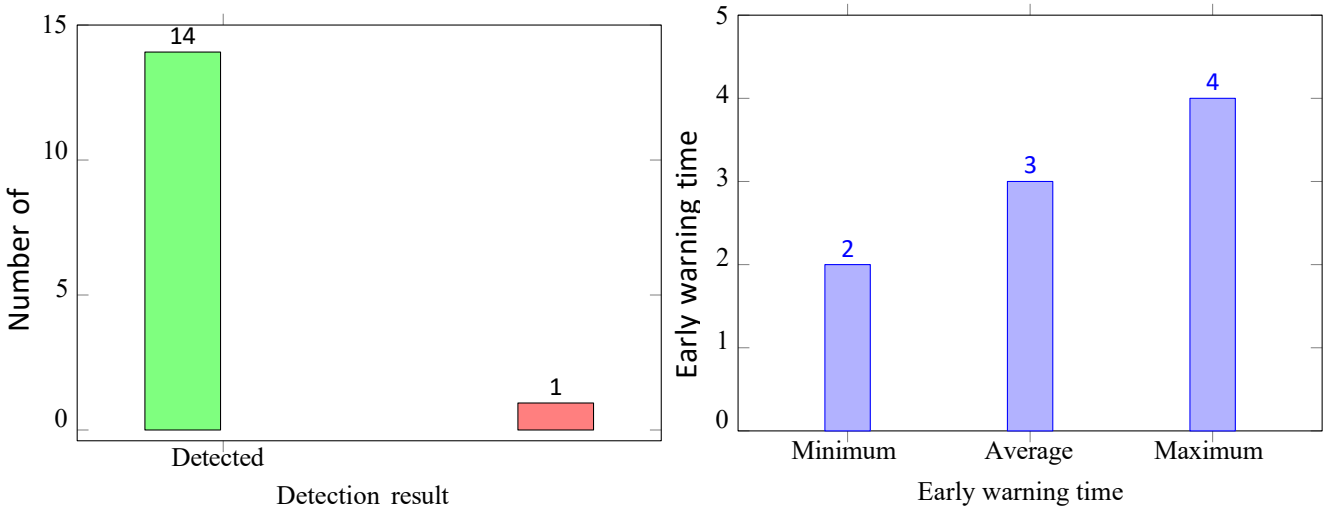
Figure 3: System block diagram

3 Machine Learning Model

A Random Forest classifier (150 trees) is used due to its robustness and interpretability. A dataset of 1,000 samples was collected over 12 weeks from multiple mountainous locations in Vietnam, covering diverse weather conditions. Features include ground acceleration (m/s^2), soil moisture (%), pore-water pressure (kPa), 24-hour rainfall (mm), temperature (°C), and humidity (%).
Samples were manually labeled into five risk levels (1–5) by geotechnical experts. Data preprocessing included outlier removal (IQR method), normalization, and interpolation of missing values. The dataset was split 70/30 for training/testing with stratified sampling. The model achieved an overall accuracy of 99%. Feature importance analysis shows soil moisture as the most influential predictor (70%), followed by pore-water pressure (15%), rainfall (10%), and acceleration (5%).

4 Experimental Evaluation

A prototype system was deployed at three test sites for 8 weeks. Field experiments indicated stable sensor performance and reliable communication. Among 15 high-risk events, the system detected 14 events (93%) with an early warning time of 2–4 hours.

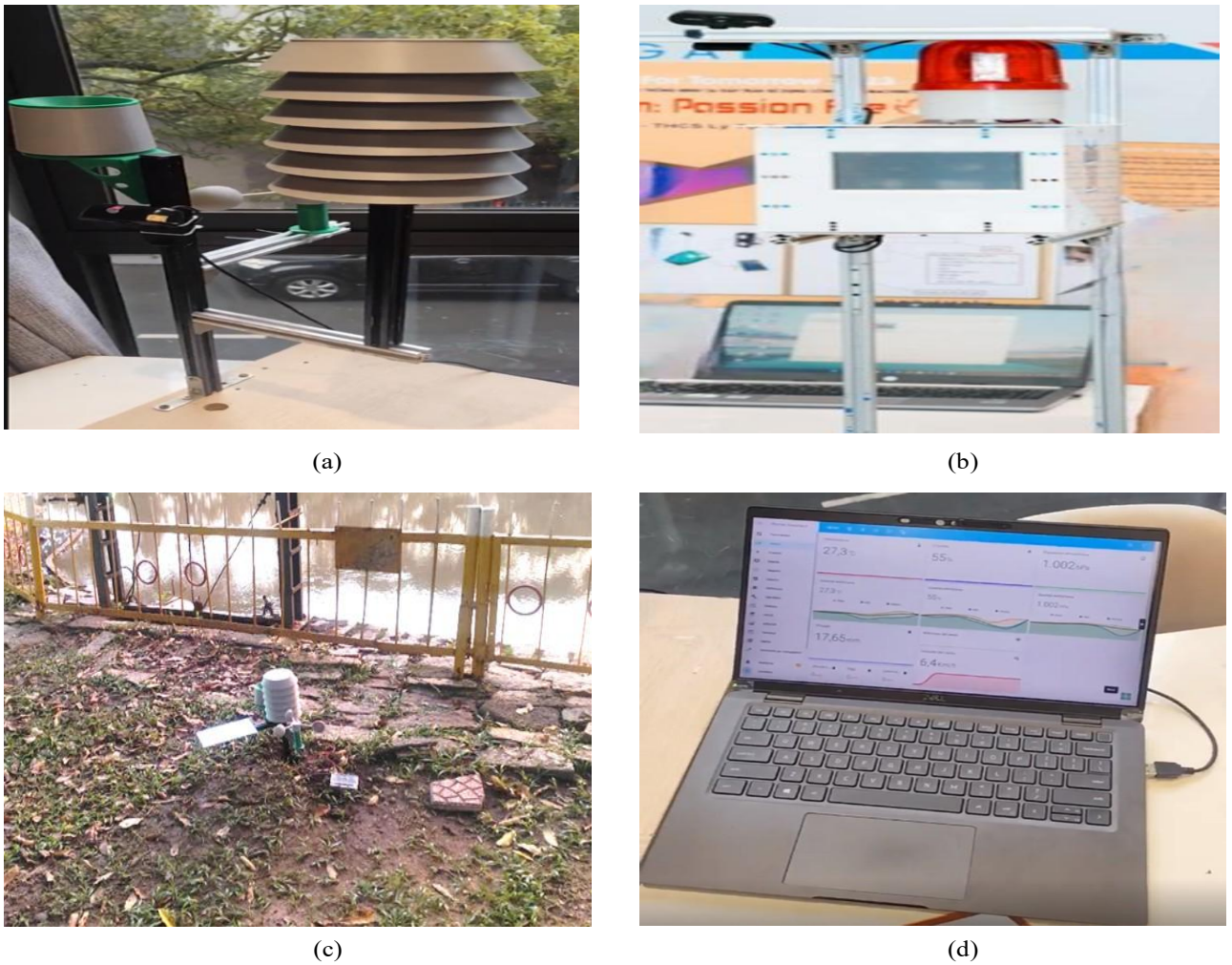


(a) Detection performance (14/15 events) (b) Early warning time

Figure 4: System performance metrics

5 Conclusion

This study presents a low-cost and high-accuracy (99%) landslide early warning system designed for mountainous communities in Vietnam. Key advantages include affordable hardware (USD 150–200), modular architecture, dual-mode alerts (local and remote), and low energy consumption. Limitations include potential sensor degradation under harsh weather conditions and the relatively small dataset size. Future work will focus on improving durability, expanding the dataset, integrating LoRaWAN, and deploying solar power solutions. The results suggest strong potential for real-world disaster risk reduction applications.



(a) (b)

(c) (d)

Figure 4: System deployment and field testing result

Acknowledgment

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